

CHE 212 : HEAT TRANSFER INNOVATION PROJECT

Forced convection through a vertical
cylinder

BY GROUP 23

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INTRODUCTION

- Definition: Convection involves heat transfer through combined conduction and fluid motion.
- Shift in Focus: Moving from buoyancy-driven "Natural Convection" to externally driven "Forced Convection" using a fan.

OBJECTIVE

- To determine the heat transfer coefficient of a vertical cylinder under forced convection using an external fan.
- To evaluate the impact of fan mounting orientation (Top, Bottom, Sideways).
- To analyze the relationship between air velocity (Low, Medium, High) and h .
- To compare experimental results with theoretical MATLAB simulations.

EXPERIMENTAL SETUP

- Core Component: Aluminium cylinder (Length: 200 mm, d: 39.86 mm).
- Heating Element: Powered by a 36V & 0.22A DC supply.
- Instrumentation: Anemometer, Digital voltmeter & ammeter
- New Addition: Variable speed fan with adjustable mounting brackets i.e., 1 for top/bottom & 2 for sideways

Modification – We lifted up the setup for the top and bottom case for better circulation of air

CONFIGURATIONS

- Top mounted fan: Downward axial flow



- Sideways mounted fan: Cross flow



- Bottom mounted fan: Upward axial flow



THEORETICAL FRAMEWORK

CONVECTION EQUATIONS

- Total Heat Input (Q): $Q = V I$.
- Newton's Law of Cooling: $Q = h A (T_s - T_{inf})$.
- Surface Area (A): $\pi * d * L$

DIMENSIONLESS GROUPS

- Grashof Number (Gr): Used for the natural convection component.
- Reynolds Number (Re): To characterize the forced airflow ($\rho * v * D / \mu$).
- Nusselt Number (Nu): Calculated via correlations (e.g., Hilpert or Nu correlation using blausis solution).
- Prandtl Number (Pr): Used in natural and forced convection's h.

GIVEN DATA

- Prandtl number(Pr)= 0.71
- $K = 0.027 \text{ W/mK}$
- $\nu = 1.7 \times 10^{-5} \text{ m}^2/\text{s}$
- $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$
- $\epsilon = 0.85$
- $g = 9.8 \text{ m/s}^2$

1. BOTTOM FAN

- $v_{\min} = 2.3 \text{ m/s}$, $T_s = 47.53^\circ\text{C}$, $T_{\inf} = 29.5^\circ\text{C}$
- $v_{\text{mid}} = 4.9 \text{ m/s}$, $T_s = 43.53^\circ\text{C}$, $T_{\inf} = 29.7^\circ\text{C}$
- $v_{\max} = 6.8 \text{ m/s}$, $T_s = 42.43^\circ\text{C}$, $T_{\inf} = 29.9^\circ\text{C}$

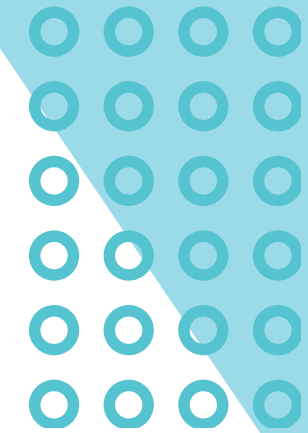
GIVEN DATA

2.TOP FAN

- $v_{\min}=5.2\text{m/s}$, $T_s=313.62\text{K}$, $T_{\inf}=302.55\text{K}$
- $v_{\text{mid}}=8.4\text{m/s}$, $T_s=312.18\text{K}$, $T_{\inf}=302.95\text{K}$
- $v_{\max}=10.4\text{m/s}$, $T_s=311.27\text{K}$, $T_{\inf}=302.75\text{K}$

3.SIDEWAYS FAN

- $v_{\min}=6.1\text{m/s}$, $T_s=39.43\text{C}$, $T_{\inf}=31.3\text{C}$
- $v_{\text{mid}}=8.4\text{m/s}$, $T_s=37.77\text{C}$, $T_{\inf}=31\text{C}$
- $v_{\max}=10.1\text{m/s}$, $T_s=37.27\text{C}$, $T_{\inf}=31\text{C}$



GIVEN DATA

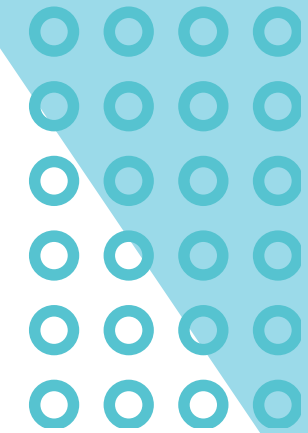
4.TOP FAN MODIFIED

- $v_{\min}=5.2\text{m/s}$, $T_s=41.9\text{C}$, $T_{\inf}=30.2\text{C}$
- $v_{\text{mid}}=7.1\text{m/s}$, $T_s=40.8\text{C}$, $T_{\inf}=30.4\text{C}$
- $v_{\max}=10.5\text{m/s}$, $T_s=40.6\text{C}$, $T_{\inf}=31.5\text{C}$

C

5.BOTTOM FAN MODIFIED

- $v_{\min}=7.2\text{m/s}$, $T_s=40.46\text{C}$, $T_{\inf}=30.2\text{C}$
- $v_{\text{mid}}=11.3\text{m/s}$, $T_s=39.16\text{C}$, $T_{\inf}=30.9\text{C}$
- $v_{\max}=12.6\text{m/s}$, $T_s=38.7\text{C}$, $T_{\inf}=30.9\text{C}$



MATLAB IMPLEMENTATION

1. TOP/BOTTOM MOUNTED FAN

- Evaluating experimental heat transfer coefficient

```
%% Experimental heat transfer coefficient
for i=1:length(Ts)
    h_exp(i)=(V*I)./((3.14*d*L).*(Ts(i)-T_inf(i)));
end
```

- Evaluating forced heat transfer coefficient

```
%% Forced heat transfer coefficient
for j=1:length(Ts)
    Re(j)=(v(j).*L)./neu;%Reynold's Number
    Nu(j)=0.664.*(Pr.^(1/3)).*(Re(j).^(1/2));
    hf(j)=(Nu(j).*K)./L;
end
```



MATLAB IMPLEMENTATION

1. TOP/BOTTOM MOUNTED FAN

- Evaluating natural heat transfer coefficient

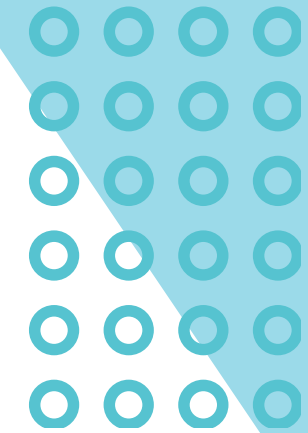
```
%% Natural heat transfer coefficient
```

```
for j=1:length(Ts)
    Tf(j)=(Ts(j)+T_inf(j))./2;
    beta(j)=1./Tf(j);
    Gr(j)=(g.*(Ts(j)-T_inf(j)).*(0.008))./((2.89*10^-10).*Tf(j));
    Nu_n(j)=0.59.*(Gr(j).*Pr).^(1/4);
    hn(j)=(Nu_n(j).*K)./L;
end
```

- Evaluating radiative heat transfer coefficient

```
%% Radiative heat transfer coefficient
```

```
for j=1:length(Ts)
    hr(j)=sigma.*eps.*(Ts(j)+T_inf(j)).*((Ts(j).^2)+T_inf(j).^2);
end
```



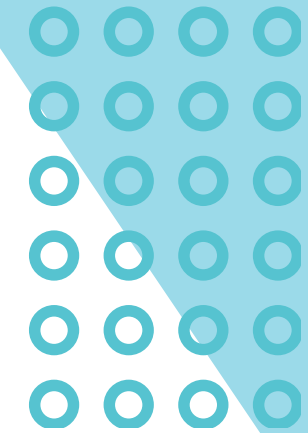
MATLAB IMPLEMENTATION

1. TOP/BOTTOM MOUNTED FAN

- Evaluating overall heat transfer coefficient

```
% Overall heat transfer coefficient & error measurement
for j=1:length(Ts)
    Nu_mix(j)=(Nu_n(j).^3+Nu(j).^3).^(1/3);
    h_conv(j)=(Nu_mix(j).*K)./L;
    h_theo(j)=hr(j)+h_conv(j);
    error(j)=(1-(h_exp(j)./h_theo(j))).*100;
end
```

- Characteristic length is taken as L because flow is along the cylinder axis
- Axial flow over the cylinder is approximated as flow over a flat plate
- Re for all three cases $< 5 \times 10^5$ i.e., laminar flow



MATLAB IMPLEMENTATION

2.SIDEWAYS MOUNTED FAN

- Evaluating experimental heat transfer coefficient

```
%% Experimental heat transfer coefficient
for i=1:length(Ts)
    h_exp(i)=(V*I)./((3.14*d*L).*(Ts(i)-T_inf(i)));
end
```

- Evaluating forced heat transfer coefficient

```
%% Forced heat transfer coefficient
for j=1:length(Ts)
    Re(j)=(v(j).*d)./neu;%Reynold's Number
    Nu(j)=0.193.*(Pr.^(1/3)).*(Re(j).^(0.618));%Hilpert correlation
    hf(j)=(Nu(j).*K)./d;
end
```



MATLAB IMPLEMENTATION

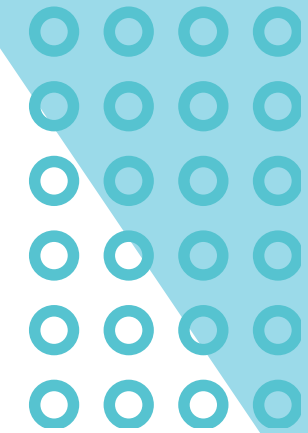
2.SIDEWAYS MOUNTED FAN

- Evaluating natural heat transfer coefficient

```
%% Natural heat tranfer coefficient
for j=1:length(Ts)
    Tf(j)=(Ts(j)+T_inf(j))./2;
    beta(j)=1./Tf(j);
    Gr(j)=(g.*(Ts(j)-T_inf(j)).*(0.008))./((2.89*10^-10).*Tf(j));
    Nu_n(j)=0.59.*(Gr(j).*Pr).^(1/4);
    hn(j)=(Nu_n(j).*K)./L;
end
```

- Evaluating radiative heat transfer coefficient

```
%% Radiation heat tranfer coefficient
for j=1:length(Ts)
    hr(j)=sigma.*eps.*(Ts(j)+T_inf(j)).*((Ts(j).^2)+T_inf(j).^2);
end
```



MATLAB IMPLEMENTATION

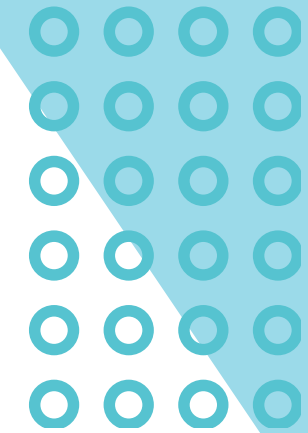
2.SIDEWAYS MOUNTED FAN

- Evaluating overall heat transfer coefficient

`%% Overall heat transfer coefficient & error measurement`

```
for j=1:length(Ts)
    Nu_mix(j)=(Nu_n(j).^3+Nu(j).^3).^(1/3);
    h_conv(j)=(Nu_mix(j).*K)./d;
    h_theo(j)=hr(j)+h_conv(j);
    error(j)=(1-(h_exp(j)./h_theo(j))).*100;
end
```

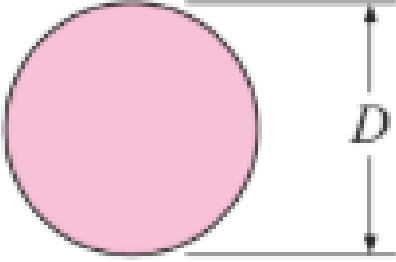
- Flow is perpendicular to the cylinder i.e., external flow over a cylinder
- Hilpert correlation is valid
- The fluid interacts with the cylinder across its diameter, so the boundary layer forms around the circumference i.e., $L_c=d$.
- Re for all three cases $< 2 \times 10^5$ i.e., laminar regime



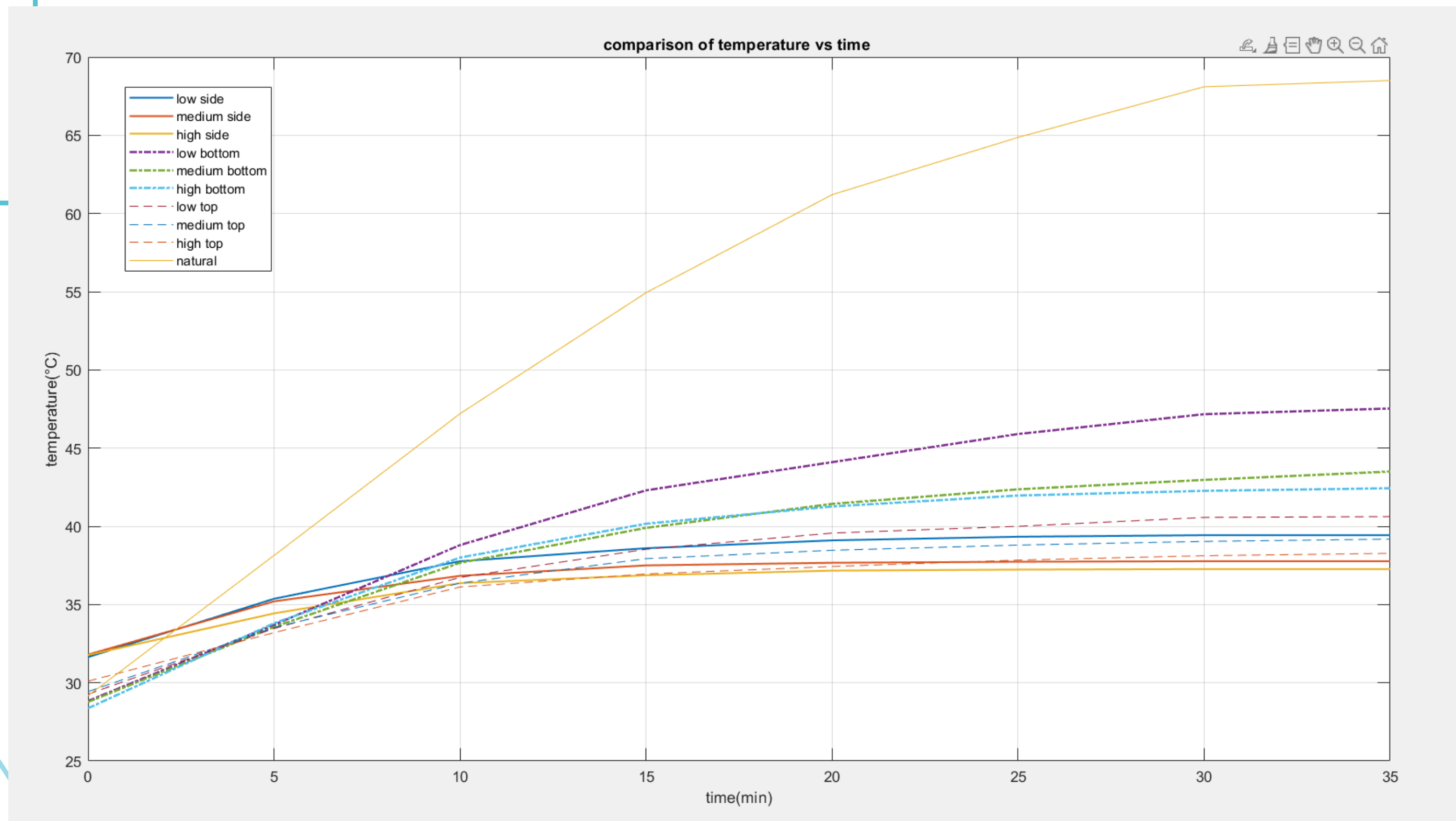
MATLAB IMPLEMENTATION

2.SIDEWAYS MOUNTED FAN

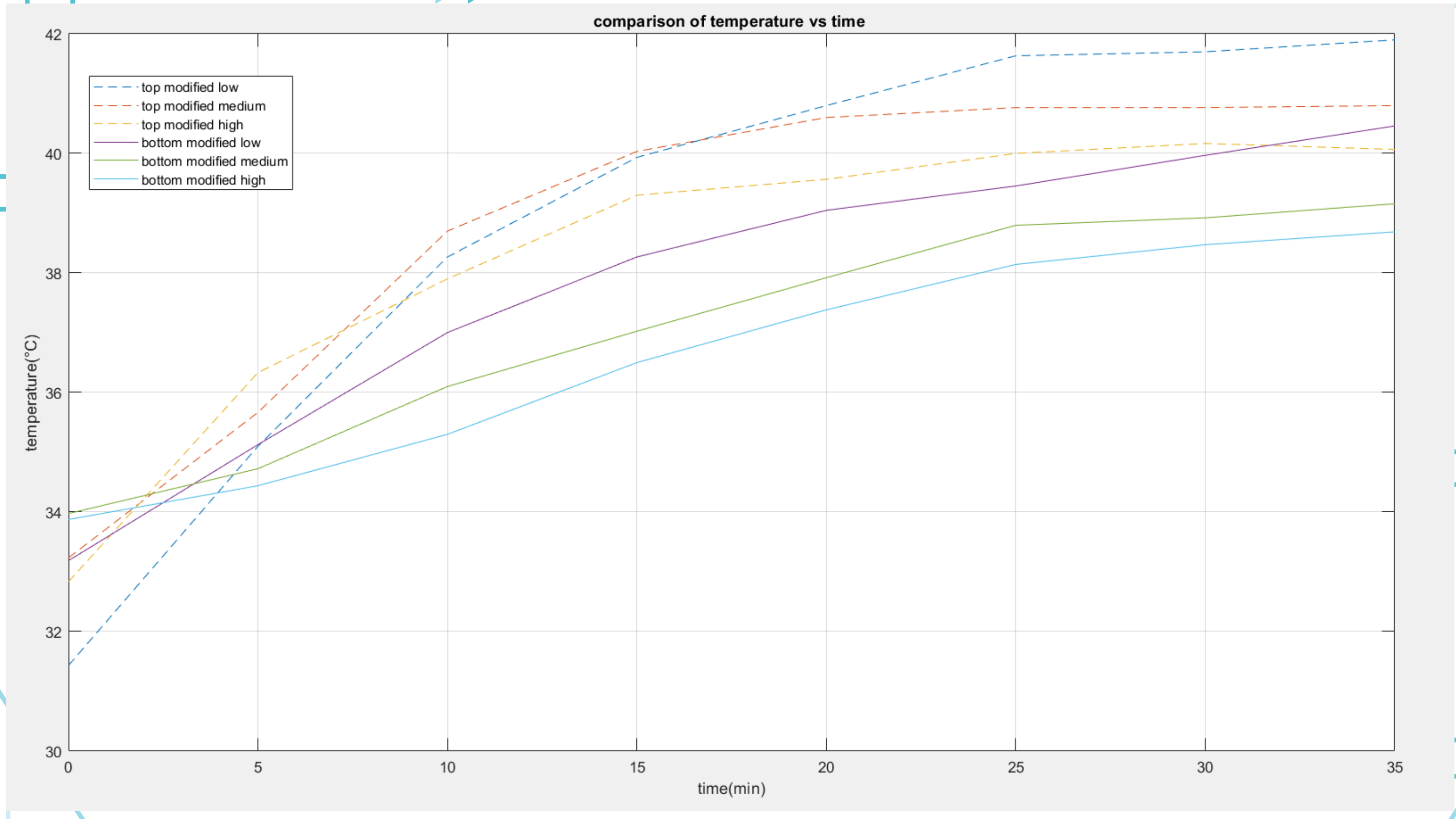
- Here we have use this table for Hilpert correlation

Cross-section of the cylinder	Fluid	Range of Re	Nusselt number
Circle 	Gas or liquid	0.4–4 4–40 40–4000 4000–40,000 40,000–400,000	$Nu = 0.989Re^{0.330} Pr^{1/3}$ $Nu = 0.911Re^{0.385} Pr^{1/3}$ $Nu = 0.683Re^{0.466} Pr^{1/3}$ $Nu = 0.193Re^{0.618} Pr^{1/3}$ $Nu = 0.027Re^{0.805} Pr^{1/3}$

TEMPERATURE VS TIME GRAPH



TEMPERATURE VS TIME GRAPH MODIFIED



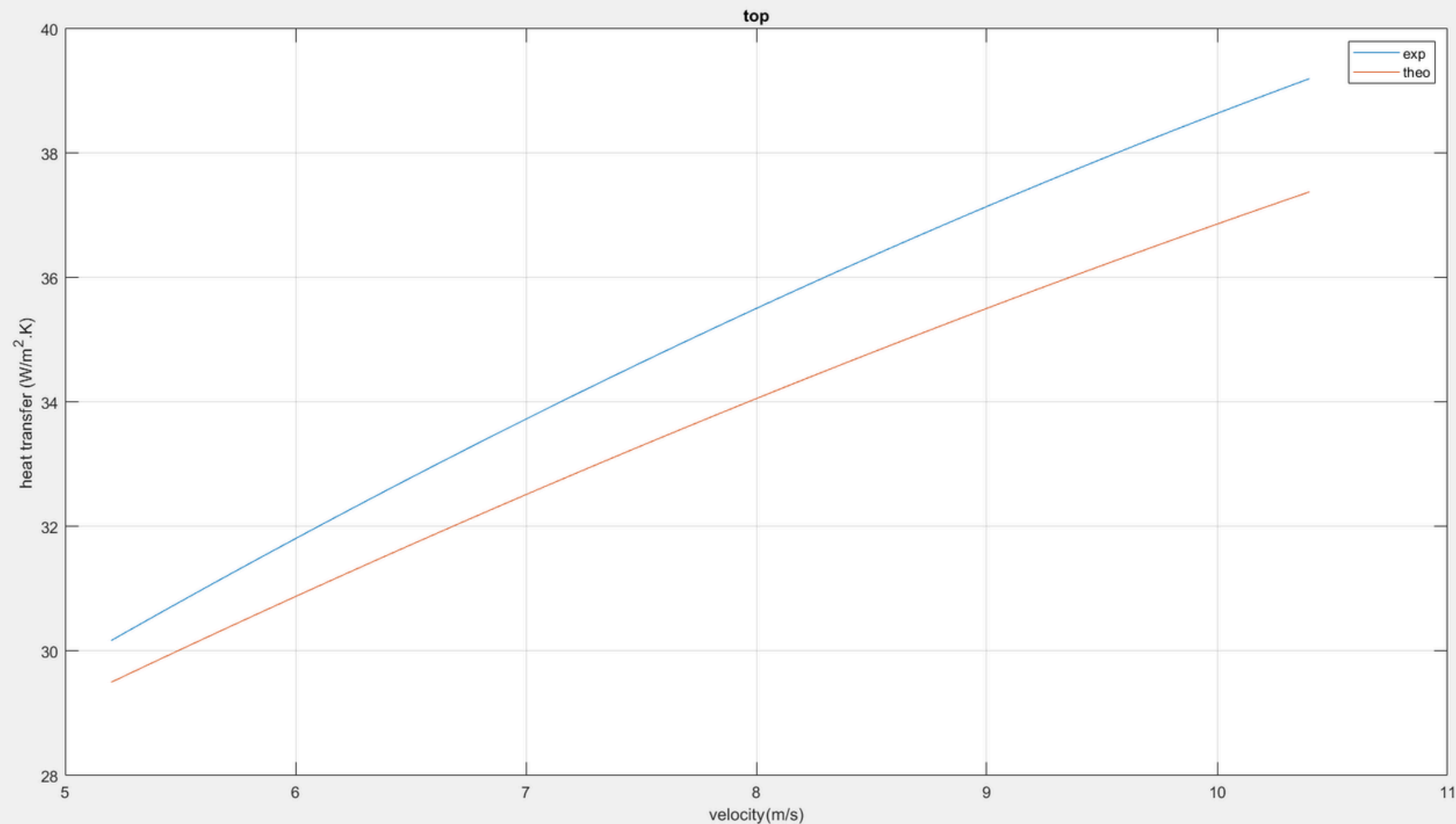
RESULTS

TEMPERATURE VS. TIME

- Natural convection's graph reached the highest temperature as no external source of heat removable.
- Bottom's graph is the higher above top and sideways as velocity is the lowest and it was flowing against the gravity so the heat removal is too low which lead to high steady state temperature.
- Top's graph is lower as the air from fan collides with surface unlike bottom & sideways, and gets back, trapped and expelled which would create turbulence there and helped to remove more heat than expected.
- Sideways graph is the lowest as it flowed crossflow and better air replacement and high heat removal lead to decrease in temperature.

RESULTS

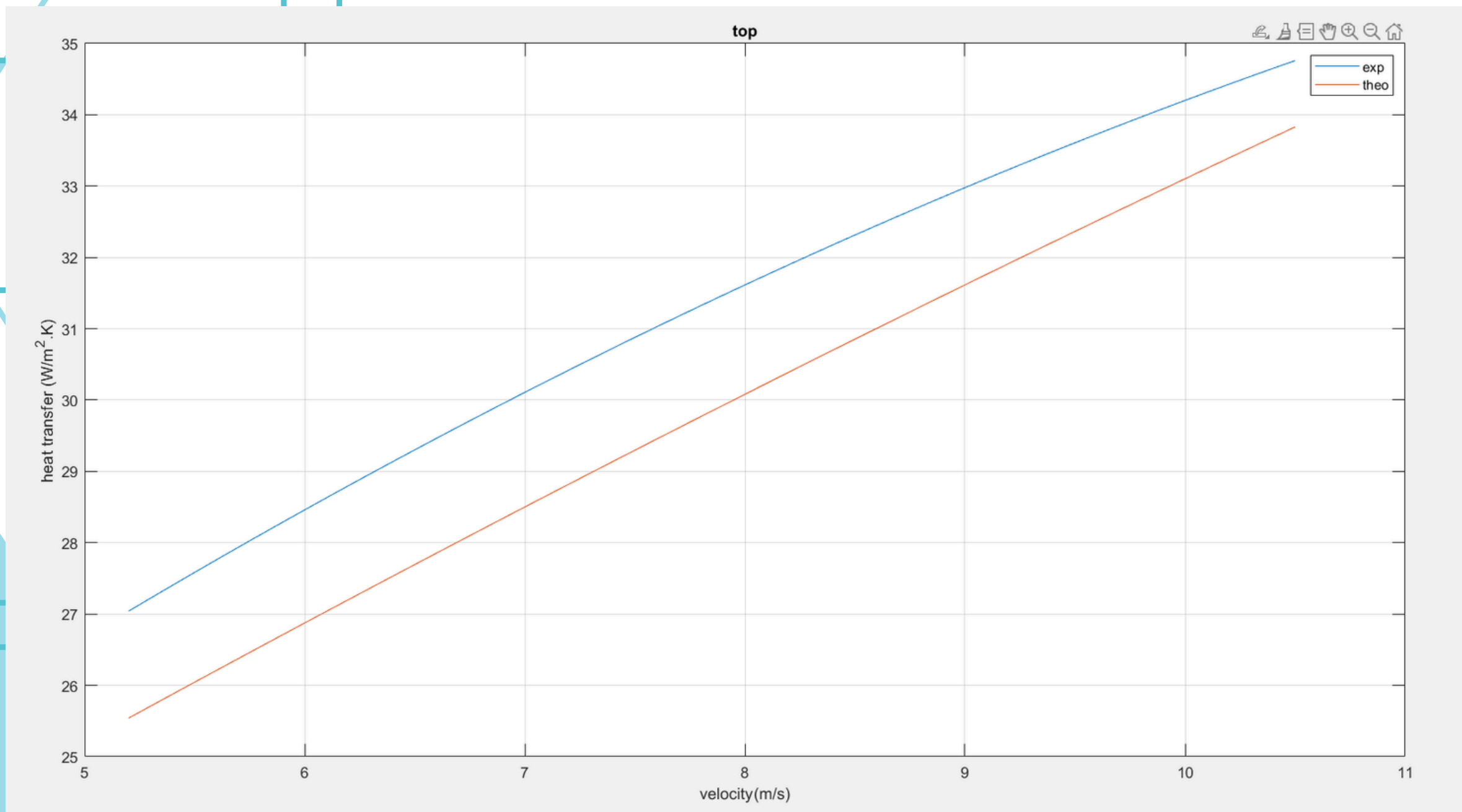
1.TOP FAN



- For this case, we get over the given velocities we got $h_{exp} > h_{theo}$ as the air got collided back and can be trapped which would increase the turbulence and increase the heat removal but in calculation we only considered the velocity measured and we eventually got less theoretical h .

RESULTS

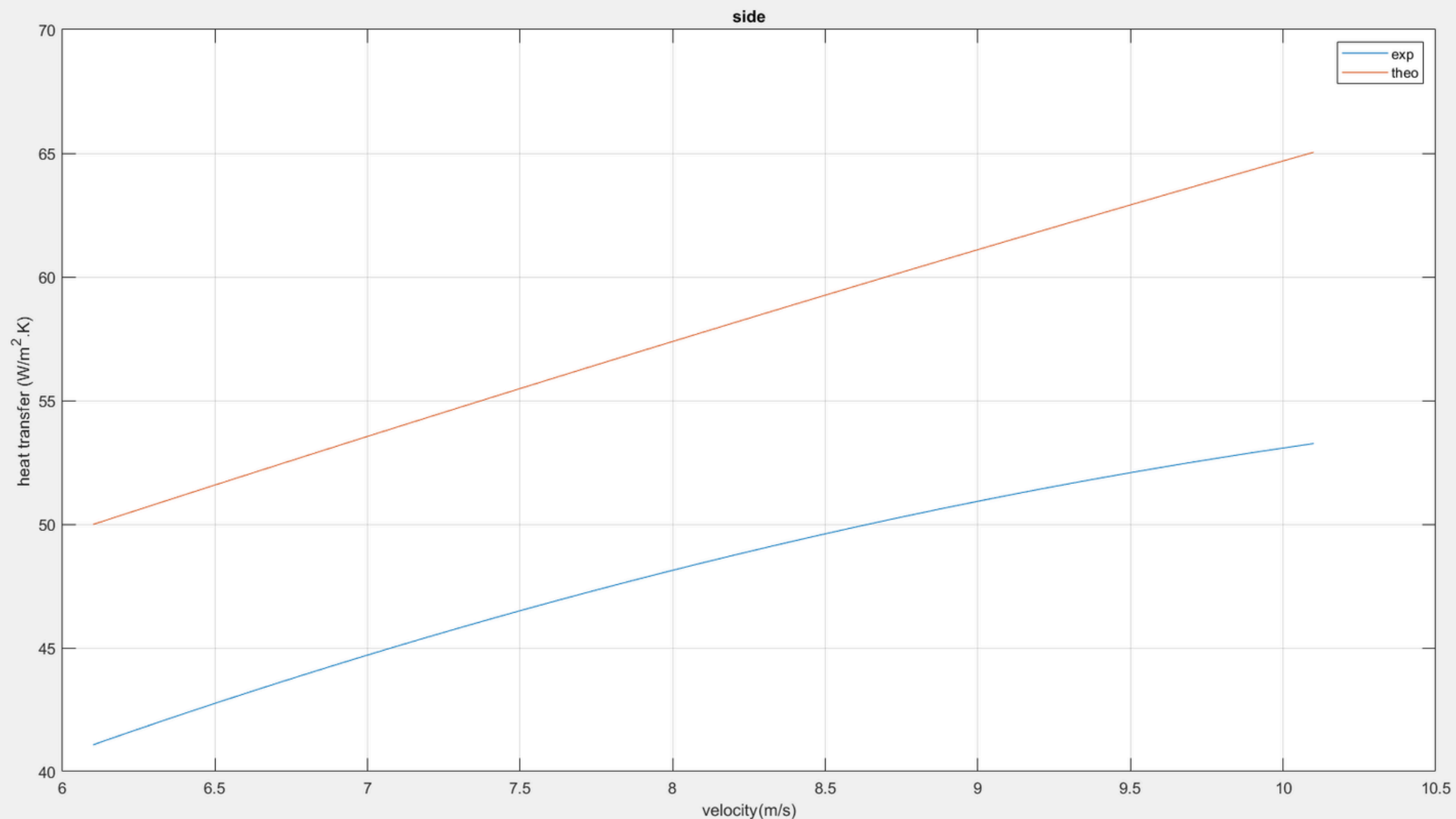
1.TOP FAN MODIFIED



- For this case, we get over the given velocities we got $h_{\text{exp}} > h_{\text{theo}}$ due to some instrumental errors i.e., we took $Q = VI$ but it's not actually like it some energy must have dissipated in resistance and all and anemometer measuring.

RESULTS

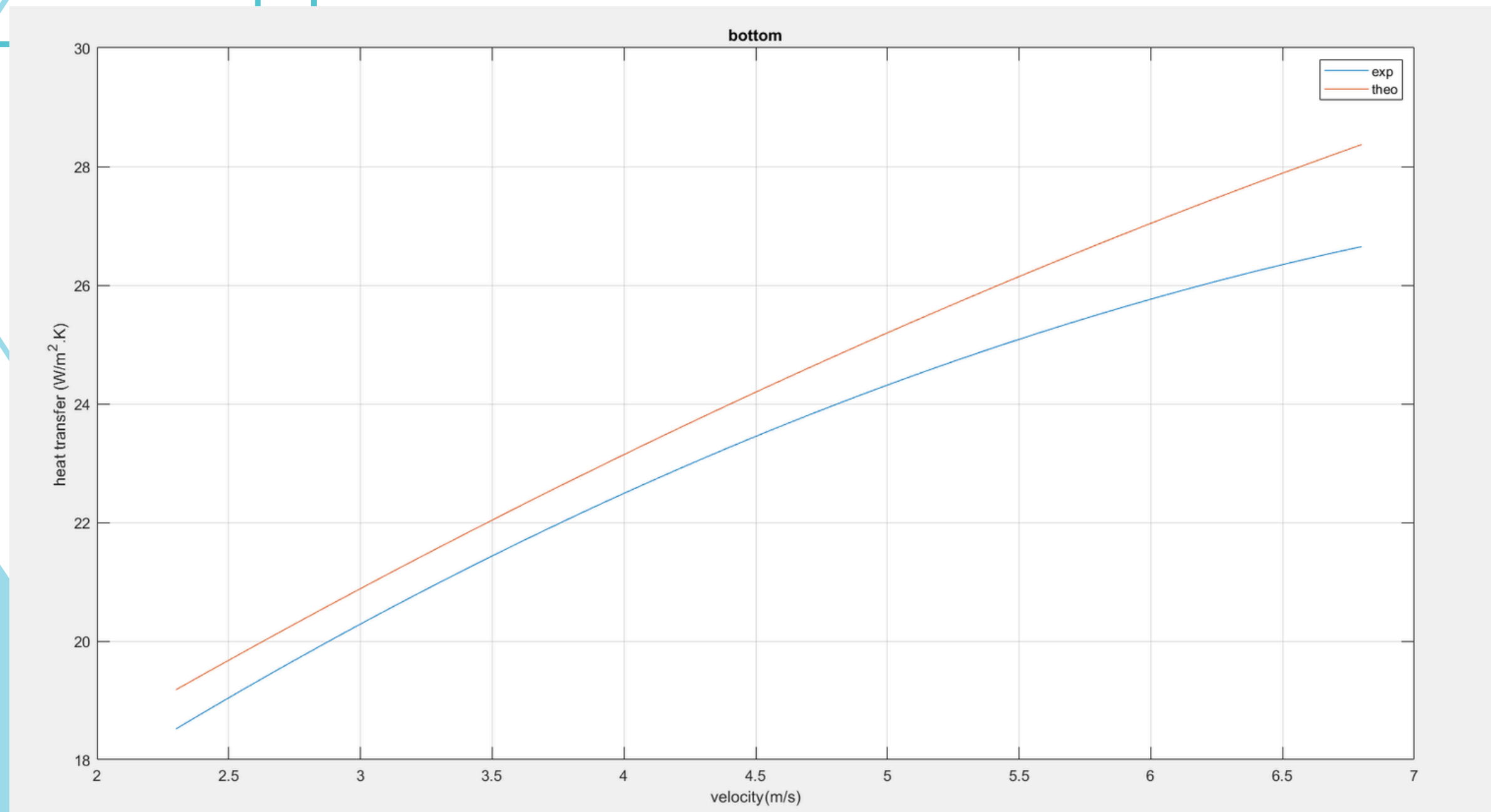
2.SIDEWAYS FAN



- For this case, we get over the given velocities we got $h_{\text{exp}} < h_{\text{theo}}$ as we took the entire area by considering heat loss from all over area would be same but the part which was towards fan would remove heat faster and against would lower which lead to theoretical h would be higher.

3. BOTTOM FAN

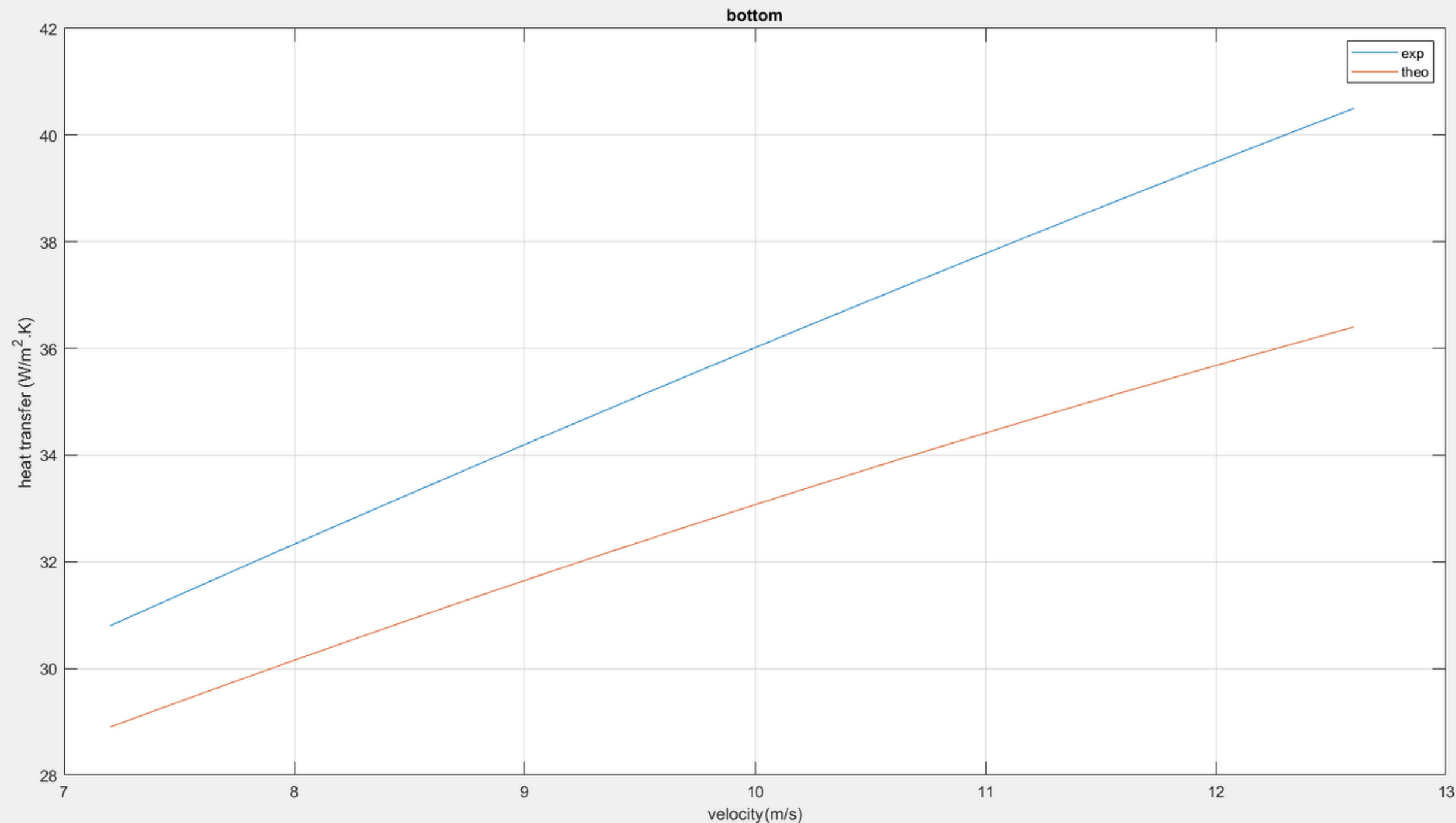
RESULTS



- For this case, we get over the given velocities we got $h_{\text{exp}} < h_{\text{theo}}$ as the heat removal is the lowest i.e., the steady state temperature be high which showed we had high radiation in this case which we did not calculated in experimental case and it gave us the experimental h lower.

RESULTS

3. BOTTOM FAN MODIFIED



- For this case, we get over the given velocities we got $h_{exp} > h_{theo}$ as we took velocity measurement with anemometer which gives error in measuring the velocity and it is obvious with increase in velocity there would be more error so $h_{exp} > h_{theo}$ and enhancing with enhance of velocity.

ERROR ANALYSIS - THEORETICAL VS. EXPERIMENTAL

1.TOP MOUNTED FLOW

- For minimum velocity,
Theoretical $h=29.5\text{W/m}^2\cdot\text{K}$
Experimental $h=30.17\text{W/m}^2\cdot\text{K}$
Error=-2.26%
- For medium velocity,
Theoretical $h=29.5\text{W/m}^2\cdot\text{K}$
Experimental $h=36.18\text{W/m}^2\cdot\text{K}$
Error=-4.4%
- For maximum velocity,
Theoretical $h=29.5\text{W/m}^2\cdot\text{K}$
Experimental $h=39.2\text{W/m}^2\cdot\text{K}$
Error=-4.9%

ERROR ANALYSIS - THEORETICAL VS. EXPERIMENTAL

1. TOP MOUNTED FLOW MODIFIED

- For minimum velocity,
Theoretical $h=25.54\text{W/m}^2\text{.K}$
Experimental $h=27.04\text{W/m}^2\text{.K}$
Error=-5.888%
- For medium velocity,
Theoretical $h=28.82\text{W/m}^2\text{.K}$
Experimental $h=30.42\text{W/m}^2\text{.K}$
Error=-5.53%
- For maximum velocity,
Theoretical $h=33.83\text{W/m}^2\text{.K}$
Experimental $h=34.77\text{W/m}^2\text{.K}$
Error=-2.77%

ERROR ANALYSIS - THEORETICAL VS. EXPERIMENTAL

2. BOTTOM MOUNTED FLOW

- For minimum velocity,
Theoretical $h=19.2\text{W/m}^2\text{.K}$
Experimental $h=18.5\text{W/m}^2\text{.K}$
Error=3.4%
- For medium velocity,
Theoretical $h=25.0\text{W/m}^2\text{.K}$
Experimental $h=24.15\text{W/m}^2\text{.K}$
Error=3.4%
- For maximum velocity,
Theoretical $h=28.4\text{W/m}^2\text{.K}$
Experimental $h=26.65\text{W/m}^2\text{.K}$
Error=6.1%

ERROR ANALYSIS - THEORETICAL VS. EXPERIMENTAL

2. BOTTOM MOUNTED FLOW MODIFIED

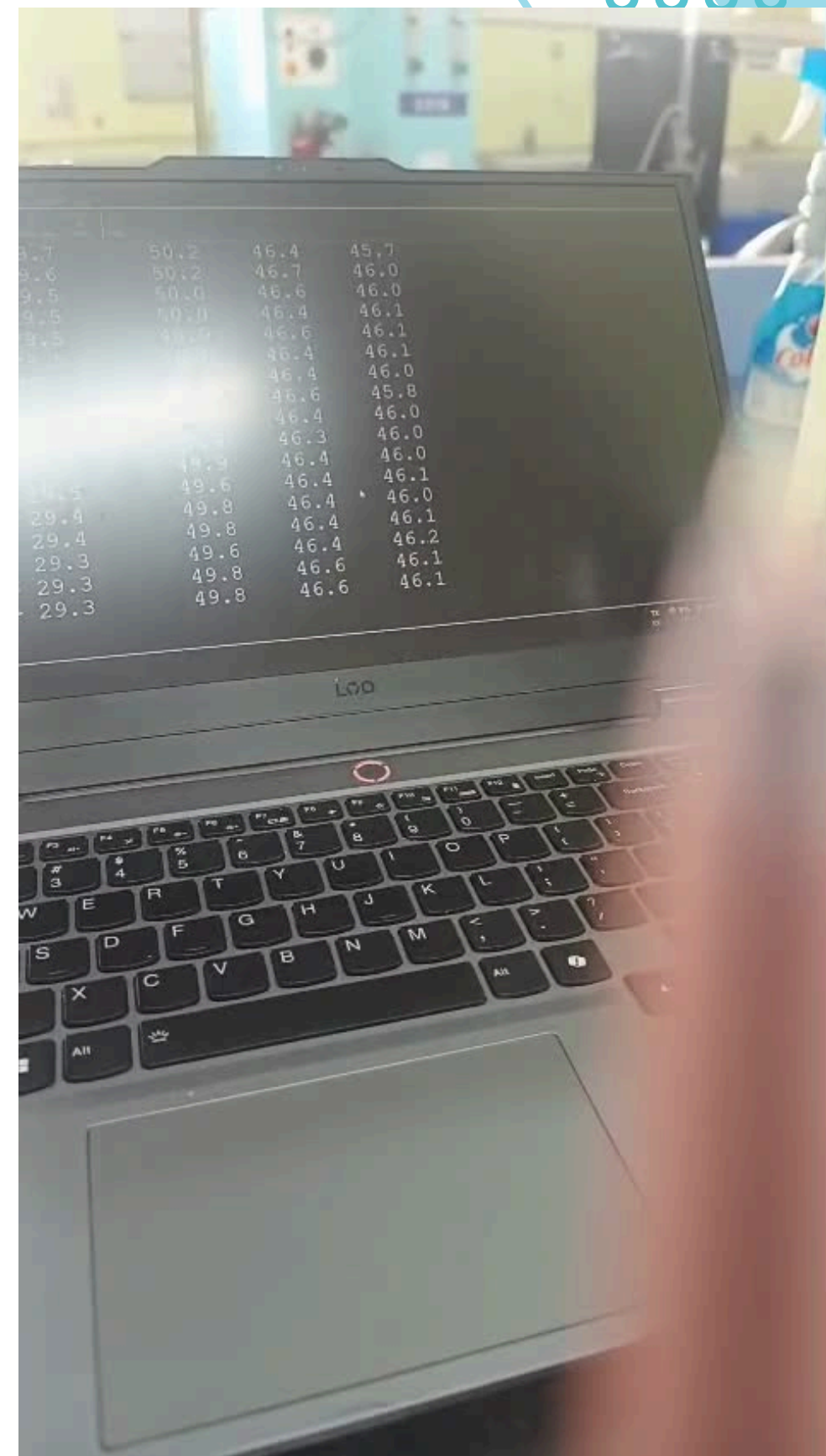
- For minimum velocity,
Theoretical $h=28.97\text{W/m}^2\cdot\text{K}$
Experimental $h=30.83\text{W/m}^2\cdot\text{K}$
Error=-6.43%
- For medium velocity,
Theoretical $h=34.82\text{W/m}^2\cdot\text{K}$
Experimental $h=38.30\text{W/m}^2\cdot\text{K}$
Error=-9.99%
- For maximum velocity,
Theoretical $h=36.44\text{W/m}^2\cdot\text{K}$
Experimental $h=40.56\text{W/m}^2\cdot\text{K}$
Error=-11.32%

ERROR ANALYSIS - THEORETICAL VS. EXPERIMENTAL

3. SIDEWAYS MOUNTED FLOW

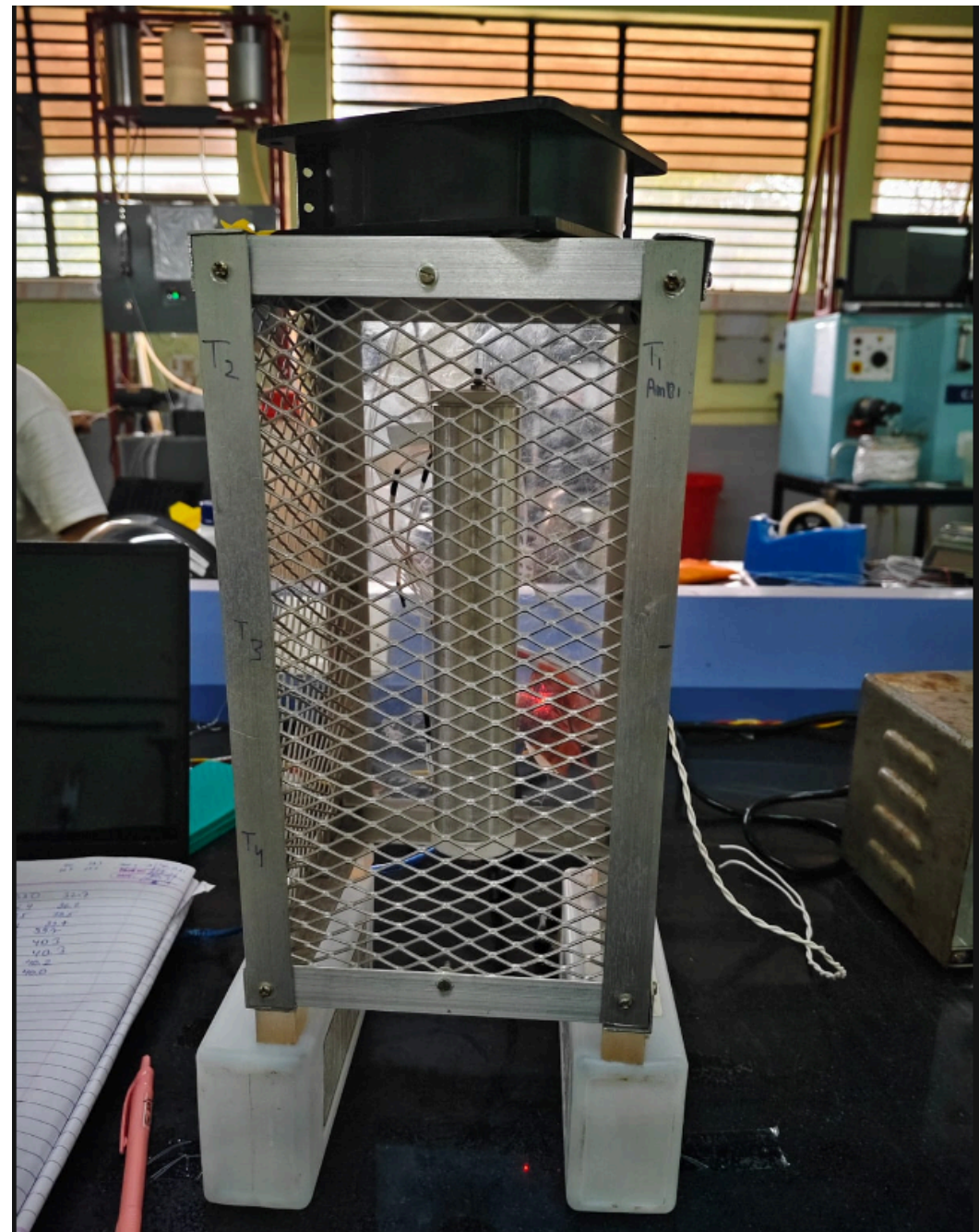
- For minimum velocity,
Theoretical $h=49.9\text{W/m}^2\cdot\text{K}$
Experimental $h=41.1\text{W/m}^2\cdot\text{K}$
Error=17.82%
- For medium velocity,
Theoretical $h=58.8\text{W/m}^2\cdot\text{K}$
Experimental $h=49.3\text{W/m}^2\cdot\text{K}$
Error=16.22%
- For maximum velocity,
Theoretical $h=65.04\text{W/m}^2\cdot\text{K}$
Experimental $h=53.3\text{W/m}^2\cdot\text{K}$
Error=18.1%

MULTIMEDIA EVIDENCE



49.7	50.2	46.4	45.7
49.6	50.2	46.7	46.0
49.5	50.0	46.6	46.0
49.5	50.0	46.4	46.1
49.5	49.9	46.6	46.1
49.5	49.9	46.4	46.1
49.5	49.9	46.4	46.0
49.5	49.9	46.6	45.8
49.5	49.9	46.4	46.0
49.5	49.9	46.3	46.0
49.5	49.9	46.4	46.0
49.5	49.9	46.4	46.1
49.5	49.9	46.4	46.0
49.5	49.9	46.4	46.1
49.5	49.9	46.4	46.2
49.5	49.9	46.6	46.1
49.5	49.9	46.6	46.1
49.5	49.9	46.6	46.1

MULTIMEDIA EVIDENCE (FOR MODIFIED)



CONCLUSION

- Successfully characterized forced convection for a vertical cylinder and heat removal for sideways would be the best among all.
- Fan orientation and velocity are critical parameters in thermal management, high velocity would give high heat transfer coefficient.

Reference: Heat and mass transfer, a practical approach by Cengel

The background features abstract geometric patterns in teal. In the top-left and bottom-left corners, there are nested rectangular outlines. In the top-right and bottom-right corners, there are clusters of small teal circles arranged in a grid. Diagonal lines cross the entire page, creating a dynamic, modern aesthetic.

THANK YOU